EXACT QRCP CAN MISS THE ORTHONORMAL-ROW CONDITIONING BOUND

SUVRIT SRA*

ABSTRACT. This note records a statement refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by exact arithmetic.

Summary: For an exact rational rank-three projector on eight coordinates, QRCP has strict pivots $I = (1, 2, 3)$ but the resulting inverse norm exceeds $\sqrt{18}$. *Certificate:* An exact rational 8×3 Grassmann-chart matrix, three positive rational pivot gaps, and the Rayleigh gap 192051/250000.

Posed by Anil Damle and Daniel Kressner (2026); counterexample found by OpenAI Codex (2026-08); formalized by OpenAI Codex; audited by OpenAI Codex; contributed by Suvrit Sra.

Let $Q \in \mathbb{R}^{n\times k}$ have orthonormal columns. Exact Businger–Golub QR with column pivoting (QRCP) applied to Q^T repeatedly chooses a remaining column with largest squared residual after orthogonal projection away from the previously chosen columns. If $P = QQ^T$ and the previously chosen indices form S, the squared residual of index i is the Schur complement

$$\rho_S(i) = P_{ii} - P_{iS}P_{SS}^{-1}P_{Si}.$$

Thus the exact pivot path depends only on the orthogonal projector P. The bound quoted below asserts that some I is well conditioned; what is at issue is whether greedy pivoting finds one. Randomized selection is the other route: dual volume sampling applied to Q^T draws I with probability proportional to $\det(Q(I,:)^TQ(I,:))$ and admits polynomial-time samplers (Li et al., 2017).

Chen, Liu, He, and Dong first answered Problem 1 negatively by an asymptotic low-coherence family (Chen et al., 2026). Their construction prescribes a long upper-triangular selected block and then completes the identity with several blocks of extra columns. The finite witness below was found later and independently. It instead specifies a small rational point in a Grassmann chart, takes its projector directly, and has only $k = 3$ and $n = 8$. No priority over Chen–Liu–He–Dong is claimed.

Problem 1 (As posed: Problem 4.3 of Amsel et al. (Amsel et al., 2026), sourced there to Anil Damle and Daniel Kressner. Chen, Liu, He, and Dong (Chen et al., 2026) were the first to publish a negative resolution.)**.** "It is known that there exists I such that $\|Q(I,:)^{-1}\|_2 \le \sqrt{k(n-k+1)}$ [HP92] and there is a fairly efficient $O(nk^2)$ algorithm that satisfies this bound [Osi25]." Problem 4.3 then asks: "Does the QRCP algorithm typically used in practice satisfy the above, or a similar, bound."

Problem 1 asks whether classical QRCP itself attains the conditioning scale known to be achievable by a suitable subset-selection algorithm (Amsel et al., 2026). The full question was first answered negatively by Chen, Liu, He, and Dong (Chen et al., 2026). Here is a different, small, exact counterexample to the displayed bound.

Date: 20 August 2026.

* TU Munich, Dept. of Mathematics, School of CIT; Garching, Germany. Email: s.sra@tum.de.

From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

Theorem 2 ([**Disproof**]: QRCP orthonormal-row bound)**.** *There are $n = 8$, $k = 3$, and $Q \in \mathbb{R}^{8\times 3}$ with $Q^TQ = I_3$ such that exact Businger–Golub QRCP applied to Q^T makes three strict pivot choices $I = (1, 2, 3)$, while*

$$\|Q(I,:)^{-1}\|_2 > \sqrt{k(n-k+1)} = \sqrt{18}.$$

Proof. Put $V = \binom{I_3}{W}$, where

$$W = \frac{1}{1000}\begin{pmatrix} 707 & -23 & -985 \\ 1639 & 1259 & -985 \\ 1639 & 1259 & -985 \\ -707 & 23 & 985 \\ 1639 & 1259 & -985 \end{pmatrix}, \qquad H = V^TV = I_3 + W^TW,$$

and define $Q = VH^{-1/2}$ using the positive-definite square root. Then $Q^TQ = I_3$, and its projector is the rational matrix $P = QQ^T = VH^{-1}V^T$. Hence all QRCP decisions can be certified using rational arithmetic alone.

For completeness, exact Schur-complement evaluation gives the following pivot residual and runner-up residual at each step:

S	pivot	$\rho_S(\text{pivot})$	$\max_{i\notin S\cup\{\text{pivot}\}} \rho_S(i)$	gap
$\varnothing$	1	$\frac{100874772187}{197275840682}$	$\frac{99899751883}{197275840682}$	$\frac{487510152}{98637920341}$
$\{1\}$	2	$\frac{4179375000}{14410681741}$	$\frac{28781505000}{100874772187}$	$\frac{474120000}{100874772187}$
$\{1,2\}$	3	$\frac{8000}{46809}$	$\frac{38809}{234045}$	$\frac{397}{78015}$

Every gap is positive, so tie handling is irrelevant and QRCP selects $I = (1, 2, 3)$.

The selected matrix is $Q(I,:) = H^{-1/2}$, so

$$\|Q(I,:)^{-1}\|_2^2 = \lambda_{\max}(H) = 1 + \lambda_{\max}(W^TW).$$

For the integer vector $x = (2, 1, -1)^T$, direct exact arithmetic gives

$$\|Wx\|_2^2 - 17\|x\|_2^2 = \frac{192051}{250000} > 0.$$

Consequently $\lambda_{\max}(W^TW) > 17$, whence $\|Q(I,:)^{-1}\|_2^2 > 18 = k(n-k+1)$. This is the claimed strict counterexample. □

Remark 3 (Scope). Problem 1 asks whether QRCP satisfies "the above, or a similar, bound". What falls here is the first half: the displayed bound $\sqrt{k(n-k+1)}$ is violated outright at $n = 8$, $k = 3$. A single finite witness says nothing about the second half, since a "similar" bound loose enough to admit this instance is not excluded by it. The asymptotic family of Chen, Liu, He, and Dong (Chen et al., 2026) is what settles the broader reading; this witness is smaller and exactly rational, not stronger.

Code and certificate

The case is the directory `counterexamples/qrcp-orthonormal-greedy/` of the archive, at commit `4a30c1c`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/qrcp-orthonormal-greedy
python verify.py
```

- Theorem 2 (result `qrcp-orthonormal-greedy`): exact arithmetic.

- `artifacts/certificate.json`: Exact rational Grassmann-chart witness, projector, QRCP residual table and strict pivot gaps, and a rational Rayleigh certificate for failure of the conditioning bound

AI USAGE

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexample here was found by OpenAI Codex (2026-08). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

ACKNOWLEDGEMENTS

The author (SS) thanks the Alexander von Humboldt (AvH) Foundation for their generous support via a Humboldt Professorship in AI.

REFERENCES

Noah Amsel, Yves Baumann, Paul Beckman, Peter Bürgisser, Chris Cama no, Tyler Chen, Edmond Chow, Anil Damle, Michal Derezinski, Mark Embree, Ethan N. Epperly, Robert Falgout, Mark Fornace, Anne Greenbaum, Chen Greif, Diana Halikias, Zhen Huang, Elias Jarlebring, Yiannis Koutis, Daniel Kressner, Rasmus Kyng, Jörg Liesen, Jackie Lok, Raphael A. Meyer, Yuji Nakatsukasa, Kate Pearce, Richard Peng, David Persson, Eliza Rebrova, Ryan Schneider, Rikhav Shah, Edgar Solomonik, Nikhil Srivastava, Alex Townsend, Robert J. Webber, and Jess Williams. Linear systems and eigenvalue problems: Open questions from a Simons workshop, 2026. Version 2.

Leheng Chen, Zihao Liu, Wanyi He, and Bin Dong. Iteris: Agentic research loops for computational mathematics. *arXiv:2606.02484*, 2026. URL https://arxiv.org/abs/2606.02484.

Chengtao Li, Stefanie Jegelka, and Suvrit Sra. Polynomial time algorithms for dual volume sampling. In *Advances in Neural Information Processing Systems 30*, 2017.